1. The size of the Bloom filter's buffer is proportional to the number of elements being inserted.
2. At the same time, the required number of hash functions only depends on the target false positive probability `p` (you can see this by substituting m* into the formula for k*: but don’t worry, we’ll discuss the math in the next section).

4.10.1 Explanation of the False-Positive Ratio Formula

In this section we’ll explain in more detail how the formulas to estimate Bloom filter’s precision are derived; first, let’s see how we obtain the estimate for the false probability ratio.

After a single element has been stored in a Bloom Filters with a capacity of `m` bits, the probability that a specific bit is set to `1` is `1/m`, and the probability that the same bit is set to `0` after all `k` bits for the element have been flipped (assuming the hash functions will always output `k` different values, for the same input[20]) is therefore

$$\left(1 - \frac{1}{m}\right)^k$$

If we consider the events of flipping any specific bit to `1` as independent events, then after inserting `n` elements, for each individual bit in the buffer the probability it’s still 0 is given by: